



Causal Machine Learning: A Deductive–Inductive Framework for Sociological Research

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Received: 28 February 2025 / Accepted: 29 January 2026
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Abstract Causal explanation is central to sociological research, shaping both theoretical development and empirical inquiry. This paper argues that causal machine learning—which integrates deductive identification strategies with inductive estimation techniques—offers an analytical approach for modeling complex, nonlinear social processes within the potential outcomes framework. We argue that causal machine learning operates through an iterative feedback loop: Theoretical assumptions guide flexible estimation, which inductively uncovers complex heterogeneities and nonlinearities, and these discoveries subsequently refine and expand sociological knowledge. Drawing on a systematic review of recent sociological research (2014–2024), we highlight how causal machine learning is advancing work in three key areas: causal effect heterogeneity, causal mediation analysis, and time-varying causal inference. These developments expand the methodological tool kit available to sociologists and strengthen the discipline’s ability to test, refine, and extend theories of social explanation. We conclude by outlining emerging directions, including high-dimensional causal inference and generative artificial intelligence, that are opening new methodological frontiers in causal machine learning for sociology.

Keywords Causal machine learning · Deductive–inductive reasoning · Causal effect heterogeneity · Causal mediation analysis · Time-varying causal inference

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Kausales maschinelles Lernen: ein deduktiv-induktiver Rahmen für die soziologische Forschung

Zusammenfassung Kausale Erklärungen stehen im Zentrum soziologischer Forschung und prägen sowohl die Theorieentwicklung als auch die empirische Analyse. Dieser Beitrag argumentiert, dass kausales maschinelles Lernen, das deduktive Identifikationsstrategien mit induktiven Schätzverfahren verbindet, einen analytischen Ansatz zur Modellierung komplexer, nichtlinearer sozialer Prozesse im Rahmen des Potenzial-Outcome-Paradigmas bietet. Wir argumentieren, dass kausales maschinelles Lernen in einem iterativen Feedbackprozess operiert: Theoretische Annahmen leiten flexible Schätzungen an, die wiederum induktiv komplexe heterogene und nichtlineare kausale Strukturen aufdecken. Diese Erkenntnisse tragen anschließend zur Präzisierung und Erweiterung soziologischen Wissens bei. Aufbauend auf einer systematischen Übersicht über neuere soziologische Forschung (2014–2024) zeigen wir, wie kausales maschinelles Lernen die Arbeiten in drei zentralen Bereichen voranbringt: der Heterogenität kausaler Effekte, der kausalen Mediationsanalyse und der zeitvarianten kausalen Inferenz. Diese Entwicklungen erweitern das methodische Toolkit der Soziologie und stärken die Fähigkeit des Faches, theoretisch fundierte Erklärungen sozialer Phänomene zu prüfen, zu präzisieren und weiterzuentwickeln. Abschließend skizzieren wir neue Perspektiven – darunter hochdimensionale kausale Inferenz und generative künstliche Intelligenz –, die weitere methodische Innovationspotenziale für kausales maschinelles Lernen in der soziologischen Forschung eröffnen.

Schlüsselwörter Kausales maschinelles Lernen · Deduktiv-induktives Schließen · Heterogenität kausaler Effekte · Kausale Mediationsanalyse · Zeitvariante kausale Inferenz

1 Introduction

Causal explanation has long been central to sociology and the social sciences more broadly, playing a key role in both the development of sociological theory and the advancement of empirical knowledge. Beyond describing social regularities, causal explanation enables sociologists to investigate underlying mechanisms, adjudicate between competing explanations, and assess the effects of specific causes and interventions (Goldthorpe 2026, this issue). This paper examines how causal machine learning—the use of machine learning methods for causal inference—can contribute to sociological explanation. We argue that integrating deductive reasoning, formalized through the potential outcomes framework, with inductive modeling via machine learning enhances sociological inquiry.

Causal inference offers a specific approach to explanation, centered on estimating the effects of causes under clearly articulated identification assumptions. These assumptions enable researchers to derive causal estimands that can, under appropriate conditions, be estimated using probabilistic models. Identification assumptions (e.g., consistency, ignorability, positivity) function as general rules under which

causal claims are made. When these assumptions are satisfied, the resulting estimands (e.g., the average treatment effect and the average treatment effect on the treated) warrant a causal interpretation. In this sense, causal inference is fundamentally deductive (Tutić and Voss 2026, this issue). Machine learning contributes to causal inference by offering flexible tools for modeling the data-generating process. Social phenomena are often nonlinear and not well approximated by simple stochastic models; machine learning methods are well suited to capturing this complexity. When theoretically informed and supported by credible identification strategies, these methods facilitate causal estimation even in high-dimensional data settings (Leitgöb and Keusch 2026, this issue).

In this paper, we synthesize sociological research from 2014 to 2024 that integrates machine learning into causal inference. Our review focuses on publications in leading general and methodological journals in sociology. We identify three key domains in which causal machine learning has made notable contributions and holds significant promise for sociology: causal effect heterogeneity, causal mediation analysis, and time-varying causal inference. For each area, we examine recent methodological developments, highlight empirical applications, and assess how machine learning has been—or could be—incorporated into causal reasoning in sociological research. We also highlight two emerging directions in causal machine learning that remain underexplored in current sociological research: causal inference with high-dimensional data and the use of generative artificial intelligence for causal reasoning. In doing so, we outline how machine learning can advance theoretically grounded causal inquiry in the next phase of sociological research. By framing causal machine learning as a deductive–inductive framework for causal inference, this paper clarifies its methodological role in supporting sociological explanation.

2 Causal Machine Learning: From Identification to Estimation

2.1 Causal Inference and Identification

Causal inference in the social sciences centers on counterfactual reasoning, with a focus on estimating the “effects of causes.” For example, we may ask about the effects of college education: What do life outcomes look like if an individual chooses to attend college versus not attending? Even with detailed data on college completion and later-life outcomes, we cannot answer this question directly because we do not observe the counterfactual: what would have happened if the same individual had made a different choice of attending versus not attending. This challenge motivates the potential outcomes framework, which formally distinguishes between observed and unobserved outcomes. Building on Neyman (1923) and Fisher (1935), Rubin (1974, 1977, 1986) defined causal effects as comparisons between potential outcomes under different treatment conditions. Let Y denote the observed outcome, A the treatment (or exposure), and X a set of pretreatment covariates that might affect the relationship between A and Y . If we let $Y(1)$ represent the potential outcome under treatment (e.g., attending college) and $Y(0)$ the potential outcome under control (e.g., not attending), the causal effect for an individual is $Y(1) - Y(0)$. However,

we only ever observe one of these outcomes. This is the “fundamental problem of causal inference” (Holland 1986), which turns causal inference into a missing-data problem.

Using the potential outcome framework, we can define a variety of causal estimands. A common estimand is the average treatment effect (ATE):

$$\underbrace{E[Y(1)] - E[Y(0)]}_{\text{Causal Estimand (ATE)}}$$

This estimand captures a core causal question: what would have happened, on average, if a randomly selected individual had received the treatment (e.g., attended college) or not received treatment? But since we never observe both $Y(1)$ and $Y(0)$ for the same individual, how can we estimate the ATE? To identify causal effects from observational data, we rely on three identification assumptions that link observed data to potential outcomes: consistency, conditional ignorability, and positivity.¹ First, the consistency assumption establishes a connection between observed outcomes and potential outcomes. It states that if a unit receives treatment level a , then the observed outcome Y is equal to the potential outcome under treatment a :

$$Y = Y(a) \text{ if } A = a$$

This implies that the potential outcomes under treatment among the treated correspond to the observed outcomes for those who actually received the treatment:

$$Y(1)|A = 1 \equiv Y|A = 1$$

In other words, the assumption of consistency allows us to identify $E[Y(1)|A = 1]$ —the average potential outcome under treatment for the treated—from observed data. To identify potential outcomes beyond the treated or untreated subgroups, we require the conditional ignorability assumption:

$$Y(a) \perp A \mid X$$

where X denotes a set of observed pretreatment covariates. This assumption implies that, within strata defined by X , treatment assignment is as good as random.² Under this assumption, we can equate the conditional distribution of potential outcomes

¹ These assumptions are standard in point-treatment settings under the potential outcomes framework. Although we do not explicitly invoke the stable unit treatment value assumption (SUTVA) in the main text, we assume it holds throughout. The SUTVA comprises two components: (1) no interference between units and (2) no hidden variations of the treatment. More complex settings—such as those involving mediation or time-varying treatments—require additional or modified assumptions (e.g., sequential ignorability).

² The covariate set X is often selected based on directed acyclic graphs (DAGs) and the backdoor criterion. For further details, see Appendix B.2.

under treatment a with the distribution of observed outcomes among those who actually received treatment a , conditional on X :

$$Y(a)|X \equiv Y|A = a, X$$

Along with the law of iterated expectations, this allows us to identify population-level potential outcomes from observed data:

$$E[Y(a)] = E_X[E[Y|A = a, X]]$$

Finally, the positivity assumption ensures that differences between potential outcomes are identifiable from observed data:

$$0 < P(A = a | X = x) < 1 \text{ for all } a \in A, x \in X$$

This means that every subgroup defined by covariates X must have a nonzero chance of receiving each treatment. Without this assumption, counterfactual comparisons such as $E[Y(1)] - E[Y(0)]$ are not meaningful because some treatment conditions are not just unobserved—they are undefined for certain subgroups. While positivity is a theoretical assumption, it also has empirical implications: In real data, violations or near-violations can occur when there is limited overlap between treated and control groups, even if the assumption is theoretically plausible.

The standard identification assumptions—consistency, conditional ignorability, and positivity—allow the ATE to be identified from the observed data distribution. This means the ATE can be expressed as a statistical quantity that depends only on observed variables.

2.2 Estimation Strategies and the Role of Machine Learning

To compute a causal effect from data, we first need a strategy to estimate the statistical quantity identified by the assumptions. The strategies generally fall into three broad categories: treatment modeling, outcome modeling, and doubly robust approaches. Treatment modeling approaches, such as propensity score matching and inverse probability weighting (IPW), model the treatment assignment mechanism, estimating $\pi_a(X) = P(A = a | X)$ and using it to reweight or match the data in a way that mimics a randomized experiment:

$$\underbrace{E[Y(1)] - E[Y(0)]}_{\text{Causal Estimand (ATE)}} = \underbrace{E\left[\frac{I(A=1)Y}{\pi_1(X)}\right] - E\left[\frac{I(A=0)Y}{\pi_0(X)}\right]}_{\text{Statistical Quantity (IPW)}}$$

This reweighting adjusts for confounding by balancing the distribution of covariates across treatment groups, making the comparison between treated and untreated units more interpretable. Machine learning can be applied here to flexibly estimate the propensity score $\pi_a(X)$, especially in high-dimensional settings in which tradi-

tional logistic regression may struggle to capture complex selection mechanisms into treatment.

We can also use outcome modeling. The g-formula provides the foundation for outcome modeling approaches such as g-computation:

$$\underbrace{E[Y(1)] - E[Y(0)]}_{\text{Causal Estimand (ATE)}} = \underbrace{\int E[Y|A=1, X=x] dF_X(x) - \int E[Y|A=0, X=x] dF_X(x)}_{\text{Statistical Quantity (g-formula)}}$$

This approach requires estimating the conditional expectation $\mu_a(X) = E[Y | A = a, X]$ and estimating $F_X(x)$, either parametrically or by using machine learning. Machine learning is especially well suited to g-computation because it aligns naturally with the outcome modeling tradition: The g-formula frames causal estimation as a problem of predicting unobserved potential outcomes. This task leverages machine learning's strengths as a flexible, high-dimensional approach.

Doubly robust estimators combine treatment and outcome models. A widely used identified function that underlies these estimators is the augmented inverse probability weighting (AIPW) expression:

$$\underbrace{E[Y(1)] - E[Y(0)]}_{\text{Causal Estimand (ATE)}} = \underbrace{E\left[\frac{I(A=1)}{\pi_1(X)} \cdot (Y - \mu_1(X)) + \mu_1(X)\right] - E\left[\frac{I(A=0)}{\pi_0(X)} \cdot (Y - \mu_0(X)) + \mu_0(X)\right]}_{\text{Statistical Quantity (AIPW)}}$$

Doubly robust estimators yield consistent estimates of causal effects if either the outcome model $\mu_a(X)$ or the treatment model $\pi_a(X)$ is correctly specified (Robins et al. 1994; Scharfstein et al. 1999; Bang and Robins 2005). This feature makes doubly robust methods particularly well suited to machine learning, which can flexibly model both components. Approaches that integrate machine learning into this framework—often referred to as debiased machine learning—also enable valid statistical inference with root- n convergence (Chernozhukov et al. 2018; Díaz 2020). This property is not guaranteed when only one component is modeled flexibly, highlighting the value of the doubly robust approach.

2.3 Causal Machine Learning as a Deductive–Inductive Paradigm

The flexibility of machine learning reflects a different modeling paradigm than that of classical statistics. Whereas traditional models typically assume a known stochastic data-generating process—such as linearity or additivity—machine learning algorithms learn patterns inductively from data, with minimal reliance on a priori assumptions (Breiman 2001). As a result, causal machine learning can yield more robust estimates in settings with high-dimensional covariates and complex or non-

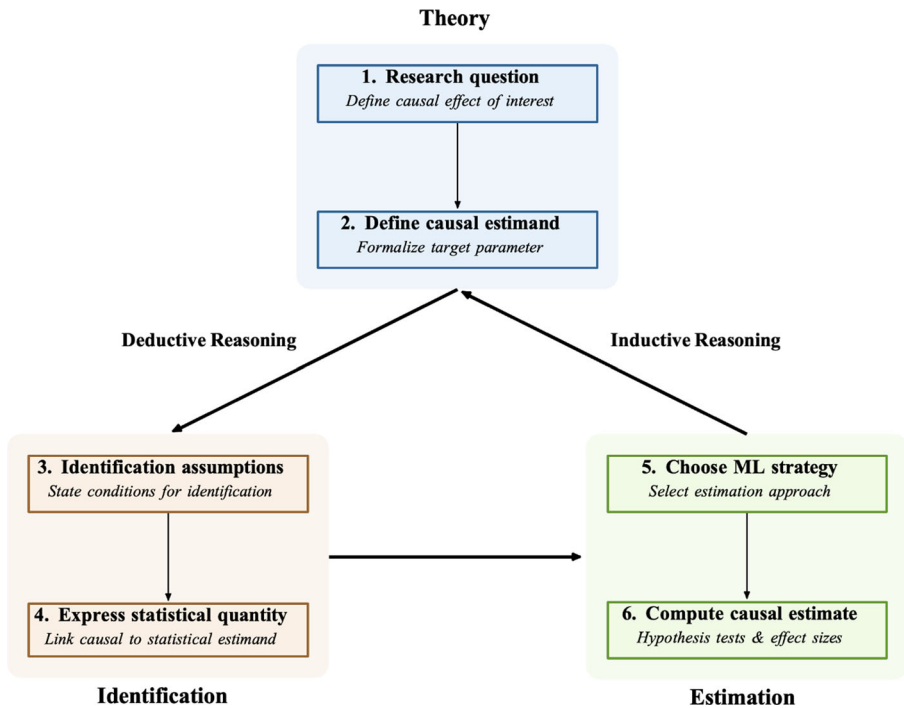


Fig. 1 Causal machine learning workflow. This diagram shows the iterative process of defining, identifying, and estimating causal effects in a triangular workflow where theory informs identification, identification guides estimation, and estimation results in feedback to refine theory

linear relationships within the covariate space—features often arising from social processes.

Yet causal machine learning is not merely an advanced estimation strategy. It is best understood as a mode of reasoning that integrates deductive design with inductive modeling. The deductive core lies in formulating a causal estimand, articulating identification assumptions, and grounding them in theory and domain knowledge. Once this causal framework is in place, the task becomes inductive: estimating the identified statistical quantity using flexible, data-driven algorithms. The causal interpretation of any estimate derives from the research design—not from the machine learning model itself. Without a clearly specified estimand and credible assumptions, even highly accurate predictions remain purely descriptive. Causal machine learning does not seek to “open the black box” but rather to embed the flexibility of machine learning within the interpretive rigor of causal inference.

As shown in Fig. 1, the causal machine learning workflow integrates deductive reasoning (used to define and identify causal estimands) with inductive modeling (used to estimate them with flexible, data-driven modeling). This integration is especially useful in settings in which researchers aim to uncover complex patterns while remaining guided by theory. For instance, a sociologist might hypothesize, based on stratification theory, that the effect of attending college differs across social groups. This hypothesis defines a heterogeneity estimand, such as the conditional average

treatment effect (CATE), whose identification depends on theoretically guided assumptions. Social theory can inform which covariates to condition on, justify the plausibility of ignorability within subgroups, and lend substantive meaning to observed patterns of effect heterogeneity. Machine learning tools such as causal forests or metalearners can then be used to estimate these effects and inductively uncover the covariates driving variation. In this way, causal machine learning supports both theory-driven inquiry and data-driven discovery.

More broadly, causal machine learning aligns closely with research practices in sociology. Sociologists routinely construct theoretical models that imply causal relationships and work with observational data to study complex, nonlinear social processes—contexts in which assuming a fixed data-generating process is often unrealistic. The causal machine learning workflow—formulating a question, justifying assumptions, and estimating flexibly—mirrors the structure of much sociological research. What machine learning contributes is the ability to scale this logic to larger, messier datasets with fewer structural constraints. Rather than replacing existing methods, causal machine learning offers an alternative approach—providing new tools to engage enduring sociological questions with greater flexibility.

This flexibility becomes even more valuable when we consider a core challenge in sociological research: Theoretical models rarely specify the precise functional forms required for parametric estimation. This is especially relevant when we assume that certain variables must be included as controls to adjust for confounding, yet we remain uncertain about the specific functional form through which these covariates affect the treatment or the outcome. Such uncertainty may reflect limited theoretical precision, but it also arises from the fundamental characteristics of social phenomena—complex systems, heterogeneous mechanisms, and the scarcity of controlled experimental settings. Causal machine learning fills this gap by embodying a deductive–inductive paradigm: It anchors inference in explicit causal assumptions while allowing the data to flexibly determine functional forms. This enables researchers to translate theoretical insights into empirically grounded models without imposing unwarranted structural constraints.

3 A Review of Causal Machine Learning in Sociology

In the preceding sections, we described the deductive–inductive logic of causal machine learning, grounded in the potential outcomes framework and enhanced by flexible machine learning methods. We next consider how these methods have been incorporated into sociological research. To assess the current state of the field, we turn to a systematic review of recent studies that integrate machine learning with causal inference.

To review the state of the art in causal machine learning in sociology, we used the Web of Science database to identify research articles published over the last decade (i.e., between 2014 and 2024). Our target journals included *American Journal of Sociology*, *American Sociological Review*, *Demography*, *European Sociological Review*, *Journal of Health and Social Behavior*, *Research in Social Stratification and Mobility*, *Social Forces*, *Social Science Research*, *Sociological Methodology*, *Soci-*

ological Methods & Research, and *Sociology of Education*—11 journals in total, selected to capture both general and methodological trends in the discipline.

We conducted a search using a combination of keywords such as “machine learning,” “causal inference,” and “causal.” We constructed queries to include terms such as “machine learning and/or causal inference,” “machine learning and/or causal,” and “causal machine learning.” We excluded tutorial-type and review papers to focus on empirical and methodological studies within sociology. Our approach ensured that the analysis focused on studies that directly contribute to the development or application of causal machine learning in sociological research.

Our search yielded 79 papers, of which 11 explicitly addressed causal machine learning. Interestingly, although many articles (68 of 79) employed causal inference frameworks, most did not frame their approach using machine learning methods. This discrepancy underscores the relatively limited—but growing—adoption of machine learning within sociological causal research.³ As shown in the top panel of Fig. 2, causal machine learning papers began to appear in sociology journals around 2020. The bottom panel shows that among the papers labeled as causal and causal & ML (causal machine learning), there is a mix of methodological and substantive work. We found a roughly comparable proportion of methodological and substantive research produced in this area since the 2020s within sociology. Appendix A, Table A.1, offers a full list of articles we reviewed. Where relevant, we also discuss working papers that have been conditionally accepted or posted on ArXiv and SocArXiv. Including these preprints reflects the rapid pace of methodological development in this area and the increasing visibility of causal machine learning in sociological discourse.

Through our literature review, we identified three major themes that reflect how causal machine learning has been incorporated into sociological research: (1) causal effect heterogeneity, (2) causal mediation analysis, and (3) time-varying causal inference. These domains are closely aligned with foundational concerns in sociological theory and method, i.e., the identification of subgroup variation in stratification research, the emphasis on mechanisms in theory testing, and the importance of understanding social processes over the life course. All three domains involve complex and dynamic social processes—challenges that machine learning is particularly well suited to model.

While our review focuses on three core domains, we also note the growing interest in emerging areas such as causal inference with high-dimensional data and the use of generative artificial intelligence for causal reasoning. Although still relatively rare in sociological publications, these developments reflect broader shifts in the computational social sciences and are likely to shape future methodological innovation in the field.

³ We also considered research from economics and political science. We narrowed the scope of the search to focus exclusively on the intersection of “causal (inference) and machine learning” and “causal machine learning.” The analysis was restricted to four prominent journals: *American Economic Review*, *American Political Science Review*, *Econometrica*, and *Political Analysis*. Using the same date range of 2014 to 2024, this more targeted search yielded 10 relevant papers. That 10 papers emerged from just four economics and political science journals—compared to 11 spread across 12 sociology journals—suggests that these fields adopted machine learning for causal inference more rapidly than sociology did during this period.

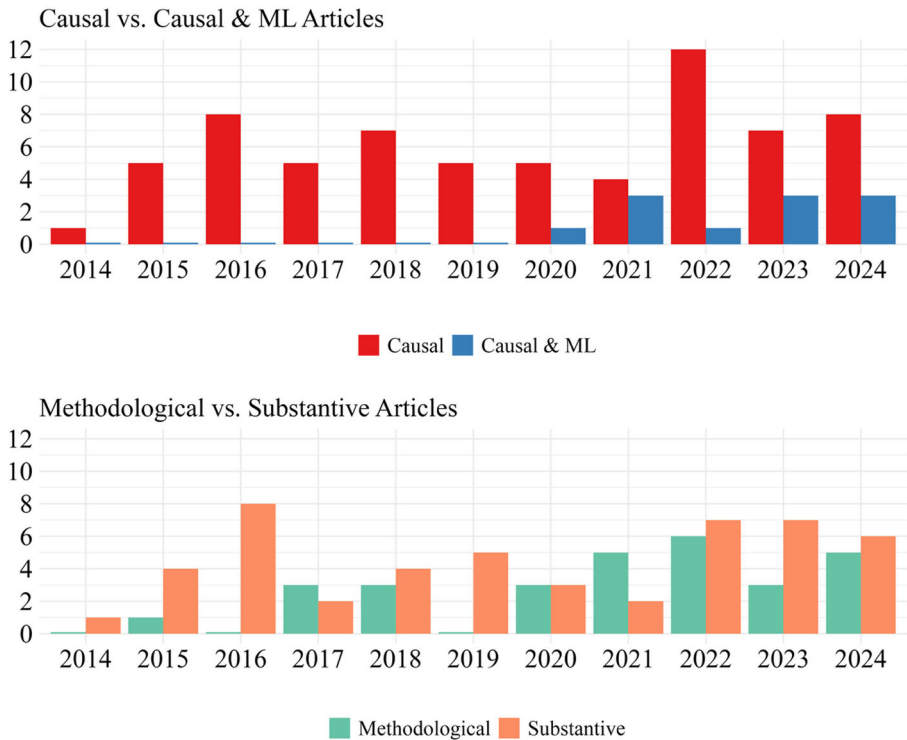


Fig. 2 Number of causal and causal machine learning articles, 2014–2024

4 Recent Developments in Causal Machine Learning in Sociology

4.1 Causal Effect Heterogeneity

In many sociological studies—whether on disadvantaged neighborhoods (e.g., Wodtke et al. 2016), college education (e.g., Brand and Xie 2010; Zhou and Pan 2023), or social policy (e.g., Daoud and Johansson 2024)—treatment effects often vary by sociodemographic characteristics or social or economic context. Classical deductive approaches typically address such heterogeneity through prespecified interaction terms or subgroup analyses. By contrast, machine learning takes an inductive approach, using flexible methods to uncover heterogeneous treatment effects and reveal complex patterns without prespecifying subgroups.

To formally investigate such heterogeneity, researchers turn to the CATE estimand.⁴ The CATE represents the difference in expected outcomes under treatment and control, conditional on a set of covariates (those with $X = x$):

$$E[Y(1)|X = x] - E[Y(0)|X = x]$$

The CATE allows for potential heterogeneity across any dimension of covariates. For example, Brand (2023) employed CATEs to demonstrate that college completion had differential effects on life outcomes, including poverty and low-wage work, depending on individuals' socioeconomic backgrounds. These findings have direct implications for both sociological theory and policy: Programs to expand college access may be more effective if targeted toward groups that benefit most.

To estimate CATEs, some sociologists have employed machine learning methods, such as causal trees and forests (Athey and Wager 2019; Brand et al. 2021; Wager and Athey 2018), as well as metalearners (Künzel et al. 2019). These models relax the linearity assumptions of traditional regressions, allowing for more flexible modeling of complex interactions. Causal trees, for instance, extend decision trees to partition data based on heterogeneity in treatment effects. Each leaf represents a subgroup with a relatively homogeneous effect, offering an interpretable summary of how heterogeneity emerges through covariate splits. For example, Brand et al. (2021) use a causal tree to show that those whose mothers had less than a high school degree, grew up in large families, and reported low social control have notably large effects of completing college on reducing low-wage work over the career.

A single causal tree can be unstable, and it may be unclear which tree best captures the underlying heterogeneity. Causal forests address this issue by averaging across many causal trees. This ensemble approach reduces variance and improves robustness while preserving the nonparametric flexibility of tree-based models. Like regression forests, causal forests split data based on covariates and treatment. From causal forests, we can generate several metrics to summarize patterns of effect heterogeneity. For example, covariate importance plots show the measures that are the strongest determinants of generating the structure of the trees in the forest, or the most important variables in determining effect heterogeneity. We can then use these variables to partition the sample into subgroups. Recent work also introduces techniques such as “distilled causal trees,” which approximate the predictions of a causal forest using a single tree fitted to the forest's estimated treatment effects (Rehill 2024). We can use the tree to identify particularly responsive subgroups (see Xu et al. [2024] for an example).

Metalearners reframe causal inference as a sequence of prediction tasks to estimate CATEs. The S-learner fits a single model with the treatment as a covariate, the T-learner estimates separate models for treated and control groups, and

⁴ The estimands—the average treatment effect (ATE), average treatment effect on the treated (ATT), and conditional average treatment effect (CATE)—are summarized in Appendix B.1, Table B.1. When treatment effects are heterogeneous, the ATT generally differs from the ATE because it averages effects over the treated subpopulation rather than the full population; the CATE provides a formal framework for characterizing this heterogeneity across covariate-defined subgroups.

the X-learner combines imputed potential outcomes to balance estimation across groups (Künzel et al. 2019). These learners can leverage any supervised machine learning algorithm—such as random forests or Bayesian additive regression trees (BART)—making them adaptable across data structures and scales.

Recent studies in sociology demonstrate how these tools can be used for studying effect heterogeneity. For example, O’Flaherty et al. (2023) used BART to estimate the effects of teen motherhood on later mental health and found largely homogeneous effects. Daoud and Johansson (2024) applied generalized random forests to examine the heterogeneous effects of International Monetary Fund–imposed austerity on child poverty. He found significant variation in effects by family wealth, with middle-class children proving unexpectedly vulnerable.

These examples underscore how machine learning can enrich sociological explanation by uncovering treatment effect heterogeneity—especially when theory offers no obvious subgroup boundaries. Causal machine learning with a focus on effect heterogeneity deepens sociological understanding of variation in the impact of social processes and helps inform more effective policy.

4.2 Causal Mediation Analysis

A central goal of sociological research is to uncover the mechanisms linking treatments and outcomes. Recent applied research in sociology shows a growing use of mediation analysis to uncover causal mechanisms. Causal mediation analysis decomposes the total effect of a treatment into a direct effect, which operates independently of the mediator, and an indirect effect, which operates through the mediator. This decomposition helps clarify the pathways through which a treatment influences an outcome and provides a basis for studying mechanisms. For example, Pedulla (2016) used a causal mediation framework to investigate the mechanisms linking nonstandard or mismatched employment histories to labor market outcomes, finding that employers’ perceptions of workers’ competence and commitment mediated these effects.

Conventional mediation analysis—typically using linear structural equation models or regression-based path analysis—relies on strong assumptions, including linearity, additivity, and limited interaction between treatment and mediator (Baron and Kenny 1986; Breen et al. 2013). Causal machine learning offers new tools to relax these assumptions, enabling flexible modeling of nonlinear mediation processes and detecting variation in pathways across subpopulations (Zhou 2024; Xu et al. 2024). This expands the scope of analyses of mechanisms.

Causal mediation analysis is formalized within the potential outcomes framework. In the potential outcomes framework, one formalization of this idea is the natural decomposition, which expresses the total effect as the sum of a direct and an indirect component. It is called “natural” because the mediator is set to the value it would naturally take under a given treatment condition. The decomposition shown below uses the mediator value under control and therefore expresses the total effect as the sum of the average direct effect (ADE) and the average causal mediation effect

(ACME). Appendix B.1, Table B.2, summarizes this and other commonly used mediation estimands.

$$\begin{aligned}
 & \underbrace{E[Y(A=1)] - E[Y(A=0)]}_{\text{Total:ATE}} = \\
 & \underbrace{(E[Y(A=1, M(A=0))] - E[Y(A=0, M(A=0))])}_{\text{Direct:ADE(1)}} \\
 & + \underbrace{(E[Y(A=0, M(A=1))] - E[Y(A=0, M(A=0))])}_{\text{Indirect:ACME(0)}}
 \end{aligned}$$

Identifying these components requires assumptions beyond standard causal inference, most notably sequential ignorability. Sequential ignorability assumes no unmeasured confounding of either the treatment–mediator or mediator–outcome relationship. When the assumptions hold, the total effect can be decomposed into direct and indirect effects.

Recent advancements in causal mediation analysis have been driven by efforts to refine estimands that more precisely capture underlying causal mechanisms. For example, Zhou (2024) models college attendance and completion as causally ordered mediators and imposes a feasibility constraint—completion can occur only if attendance has occurred—clarifying the decomposition of effects. Encoding this constraint allows the exclusion of implausible counterfactuals (e.g., graduating from college without attending), simplifying identification and reducing reliance on parametric assumptions. Building on this, Opacic (2025) developed a method to model educational progression as a monotonic sequence of mediators, enabling estimation of path-specific effects on later-life outcomes.

Zhou and Pan (2023) illustrate the empirical potential of these methods in their study of Black–White earnings inequality. Using a simplified estimand and a debiased machine learning estimator developed by Zhou (2024), they estimated the extent to which college completion mediated the effect of college attendance. Their results suggest that degrees reduce the unexplained earnings penalty for Black men, although completion disparities limit their equalizing potential.

While these studies refine estimands to better reflect causal processes, they often assume that mediation effects are constant within prespecified subgroups. To address this limitation, Xu et al. (2024) introduced causal mediation forests—a machine learning approach that inductively detects heterogeneity in mediation pathways across covariate profiles. The method recursively partitions the data based on differences in estimated effects, eliminating the need for analysts to define subgroups in advance. They considered the effects of college attendance mediated through completion on reducing time spent in poverty and how those effects differed across the population.

These recent advances in mechanism–focused causal analysis, particularly those that refine estimands to better capture causal pathways, have inspired new approaches to studying group-based disparities in sociology (Lundberg 2024; Opacic et al. 2025; Yu and Elwert 2025). While mediation analysis typically focuses on how a treatment operates through a mediator for individuals, decomposition methods aim to

explain observed disparities between social groups. Both approaches share a common goal: uncovering the causal mechanisms through which causes shape outcomes. For example, Yu and Elwert (2025) developed a nonparametric causal decomposition framework that breaks down group disparities into three causal components: treatment prevalence, treatment effects, and differential selection into treatment. These correspond to distinct causal pathways through which group inequality emerges, clarifying how specific mechanisms contribute to observed group disparities.

4.3 Time-Varying Causal Inference

From a life-course perspective, social and developmental processes unfold over time, with earlier exposures often shaping later outcomes (Elder 1985). The timing, duration, and sequencing of interventions can significantly impact life trajectories. Treatment effects may differ depending on when an exposure occurs (e.g., early vs. late; Elías Alvarado 2016; Hicks et al. 2018), how long it lasts (short-term vs. sustained; Wodtke et al. 2022), and whether it interacts with prior conditions (Zhao et al. 2021).

To formalize these ideas, we consider a trajectory-level causal estimand that captures the total effect of following an entire sequence of treatments rather than a single point intervention. Specifically, we focus on the ATE of a treatment trajectory,

$$E[Y(\vec{a}^{(1)})] - E[Y(\vec{a}^{(0)})]$$

where $\vec{a}^{(1)}$ denotes a sequence of treatment interventions over time and $\vec{a}^{(0)}$ denotes a reference or control trajectory, often corresponding to the natural course with no intervention. This estimand represents the expected difference in potential outcomes between following the treatment sequence and the control sequence, and it is particularly well suited to longitudinal settings in which treatments are administered repeatedly and effects accumulate over the life course (Appendix B.1, Table B.3).

Time-varying causal inference is challenging because treatments and confounders interact and evolve in a feedback loop over time. Under the potential outcomes framework, sociologists have developed methods to address these complexities. Brand and Xie (2007) emphasize that time-varying causal inference requires consideration not just of two potential outcomes but of a matrix of potential outcomes corresponding to treatment histories. They propose a weighted composite estimator that accounts for this complexity. Elwert and Pfeffer (2022) suggest using future treatments as proxies for unmeasured confounders, based on the idea that confounders influencing earlier treatments also likely affect later ones. Similarly, Dafoe (2018) formalizes the assumptions required for identification under temporal dependence and provides guidance on when to condition on lagged outcomes.

To meet these methodological demands, researchers have developed a family of tools known as g-methods—including g-computation, marginal structural models (MSMs), and structural nested mean models (SNMMs). These approaches explicitly account for feedback between treatments and confounders over time, enabling estimation of total effects of treatment trajectories—such as patterns of life stability, life histories, or the timing of key life events. Unlike traditional regression models,

g-methods avoid bias from time-varying confounders affected by prior treatment. Each method adopts a distinct strategy: g-computation models potential outcomes directly, MSMs use weighting to adjust for confounding, and SNMMs estimate contrasts between nested counterfactuals within individuals. For more details on each method, see Hernán and Robins (2020) and Naimi et al. (2017).

Among these approaches, MSMs are the most widely used in sociology. They employ inverse probability of treatment weights (IPTWs) to create a pseudopopulation in which treatment is unconfounded by prior covariates and treatment histories. This enables the use of standard regression methods while adjusting for time-varying confounding. For example, Perkins (2019) used MSMs to study the effects of household instability on educational outcomes. Similarly, Lim and Wiertz (2024) examined how a history of religious involvement affects volunteering in adulthood. Lundberg et al. (2024) extend MSMs to model the effects of volatile life-course trajectories.

While MSMs offer accessibility and broad software support, they are limited in flexibility. They require accurate estimation of the treatment assignment mechanism at each time point, and their parametric specifications for marginal models may fail to capture complex, nonlinear interactions or temporal dependencies. In contrast, g-computation is particularly well suited for integration with machine learning, which can help relax these assumptions. Deep learning-based g-computation, in particular, can flexibly model time-dependent confounding by leveraging neural architectures designed for sequential data for natural language processing tasks.

Recent advances in deep learning have made g-computation feasible for high-dimensional, longitudinal data. Recurrent neural networks and transformer architectures are especially effective for modeling time-dependent processes (Koch et al. 2025; Frauen et al. 2023; Hess et al. 2024; Li et al. 2021; Xiong et al. 2024). Recurrent neural networks capture sequential dependencies in treatment and covariate histories (e.g., G-Net, 2021), and transformers flexibly attend to nonlocal, variable-length dependencies across time steps (e.g., G-Transformer, 2024). These architectures enable the simulation of counterfactual treatment trajectories under nonlinearity and treatment effect heterogeneity.

Thus far, most deep learning applications have focused on individual-level effects, dynamic treatment regimes, and repeated outcomes (e.g., clinical events or symptoms). A dynamic treatment regime is a sequence of personalized decision rules that tailors an intervention to an individual's evolving needs and history to optimize their long-term outcome. In contrast, sociology is often concerned with population-level effects on life outcomes, such as long-term earnings or later-life health. These outcomes result from socially structured trajectories rather than from the individualized policy regimes common in medicine. To address this gap, Jeon (2025) developed a transformer-based g-computation framework tailored to sociological goals. By simulating potential outcomes under typical life course trajectories, the approach estimates average effects of time-varying exposures at the population level.

5 Emerging Directions in Causal Machine Learning in Sociology

Although causal machine learning is gaining traction in sociology, its scope remains limited. As indicated by the relatively small number of papers applying these methods, we encourage more researchers to adopt the methods we describe above. But there are also two areas in particular that warrant greater attention moving forward: (1) causal inference with high-dimensional data and (2) the emerging role of generative models, including large language models (LLMs) for sociological research.

5.1 Causal Inference with High-Dimensional Data

Causal inference with high-dimensional data is an area in which machine learning's comparative advantage is especially apparent. Machine learning-based estimators are well suited to settings with many covariates, nonlinear interactions, or latent structures—conditions that often overwhelm traditional parametric models. Yet these approaches remain relatively rare in sociology, with some notable exceptions.

One promising stream of work involves integrating unstructured text into causal frameworks. For example, Duxbury (2023) used over one million news articles to estimate how racial threat narratives mediate the relationship between homicide and incarceration. Drawing on supervised machine learning to classify articles into threat-related frames, the study embedded high-dimensional text features within a causal mediation analysis. Related applications are emerging across the social sciences. For example, studies by Egami et al. (2022) and Feder et al. (2022) highlight how latent representations of texts (e.g., embeddings, topic proportions) can function as treatments or outcomes in causal analysis. These advances illustrate the compatibility of deductive causal reasoning with inductively learned representations, a methodological synthesis that expands the scope of causal inference.

A second frontier involves digital behavioral data (DBD), such as traces from social media platforms, mobile apps, or digital sensors, for causal inference (e.g., Mehrotra et al. 2017; Nyhan et al. 2023). These data are also high-dimensional and temporally rich, offering unprecedented granularity for studying social processes. For example, Mehrotra et al. (2017) estimated the causal impact of using specific mobile applications reflecting external user contexts such as socializing and traveling on self-reported happiness and stress levels. However, unlike surveys or administrative records, DBD are found data: byproducts of commercial platforms and digital activities rather than datasets intentionally designed to be representative. This quality of the data raises methodological concerns about bias, selection, and external validity (Breen and Pan 2026, this issue). In this special issue, Leitgöb and Keusch (2026) outline a design-based framework for integrating DBD into causal inference. As with text data, the key is not simply to model complexity but to embed high-dimensional inputs within theoretically grounded identification strategies.

5.2 Generative AI as a Tool for Causal Inference

Large language models are generative tools based on decoder-only transformer architectures, trained to model text sequences by predicting the next token (Vaswani

et al. 2017; Brown et al. 2020). While their outputs reflect statistical associations in massive training corpora, LLMs can generate coherent, context-sensitive responses across diverse inputs. These capabilities have opened new possibilities for sociological research (Davidson and Karell 2025); LLMs have been used to impute missing survey data (Kim and Lee 2023) and classify job descriptions into occupational categories (Laughlin et al. 2025). They have also been fine-tuned on panel survey data, such as the Panel Study of Income Dynamics and the National Longitudinal Surveys, by converting tabular records into resume-like text to predict individuals' next occupations (Athey et al. 2024).

Large language models can support causal research when embedded in designs that define clear estimands and satisfy identification assumptions. One promising direction is using LLMs to generate experimental stimuli, such as text-based vignettes, survey prompts, or visual treatments, for randomized experiments (Imai and Nakamura 2024; Zhang 2025). For example, Zhang (2025) has demonstrated how LLM-guided image editing can produce realistic visual stimuli that vary along experimentally controlled dimensions. This approach enhances experimental control, expands the diversity and realism of stimuli, and enables scaling studies beyond the limitations of handcrafted materials. It also opens new possibilities for testing sociological theories of social meaning and evaluation in a reproducible, scalable, and theory-driven way.

A second approach treats LLMs themselves as experimental agents for studying algorithmic bias and its social and policy implications. In a growing number of audit studies, LLMs have been exposed to randomly assigned vignettes such as résumés with race- or gender-coded names (Armstrong et al. 2024), clinical cases with varying patient demographics (Pendse et al. 2025), and social media posts differing in slurs and author identity (Davidson 2025) and their outputs analyzed for algorithmic bias. Davidson (2025), for example, conducted a randomized audit of multimodal LLMs and found persistent racial and topical biases, revealing important limitations in their use for moderating hate speech. Crucially, it is the causal design of random assignment and clearly defined contrasts that enable such assessment. Without it, differences in outputs may reflect uncontrolled confounding.

Looking ahead, combining generative models with theory-driven causal inference may help bridge the gap between data-rich modeling and sociological explanation. These tools expand what is possible, but they do not replace the need for rigorous research design and interpretive clarity.

6 Conclusion

This paper has examined how causal machine learning is being applied in sociological research across three key domains: causal effect heterogeneity, causal mediation analysis, and time-varying causal inference. In each area, machine learning expands the capacity of existing causal inference strategies by modeling nonlinear, dynamic, and socially structured processes. For studying effect heterogeneity, methods such as causal forests and metalearners uncover variation in treatment effects that would be difficult to prespecify in conventional regression models. In causal

mediation analysis, recent innovations include refining causal estimands to target specific mechanisms and using machine learning estimators to uncover heterogeneous mediating processes. In longitudinal settings, deep learning architectures are opening new avenues for life-course research by estimating the effects of typical life trajectories.

Taken together, these developments demonstrate how causal machine learning can expand the scope of causal inference in sociology. Importantly, causal machine learning complements rather than replaces the deductive logic of causal inference. At its core, causal inference depends on clearly defined estimands and identification assumptions that allow researchers to interpret estimates as causal effects. Once these assumptions are articulated, machine learning provides flexible tools for estimation. In this way, causal machine learning operates within a deductive–inductive paradigm: Deductive reasoning specifies the causal questions and identification conditions, while inductive modeling uncovers functional forms, interactions, and contextual variation without restrictive parametric assumptions. In some cases, this leads to further refinement of sociological theory. This synergy is especially well suited to sociological inquiry, where theories often engage contingent processes shaped by the intersection of agency and structure, i.e., patterns that resist oversimplified modeling assumptions.

Causal machine learning is well aligned with causal scholars' tradition of studying the "effects of causes." Sociologists who use causal methods define causes first and hypothesize their effects through theoretical reasoning. This makes reproducibility especially important: Results speak only to the specific causes and identification assumptions articulated by researchers. To build a robust sociological explanation, the field requires a critical mass of studies that can be compared and synthesized. As Goldthorpe (2026) notes in this issue, single estimates of the "effects of causes" often face limits to external validity, since causal claims are bounded by the contexts in which they are identified. In the presence of effect heterogeneity, the estimates may differ widely depending on the context. From this perspective, cumulative explanation emerges through replication across diverse settings and recognition of effect heterogeneity. Reproducibility thus serves not only the goal of transparency but also the collective project of building causal knowledge of the complexity of social life (Diekmann 2026, this issue).

Despite its promise, causal machine learning has yet to see widespread use in sociology. Conceptually, machine learning is often equated with prediction tasks and seen as incompatible with explanation and theory-building—a misconception this paper seeks to address. In practice, however, machine learning's primary contribution lies not in deciding what to estimate but in flexibly estimating well-defined causal effects once the estimands have been specified through deductive reasoning. Beyond these conceptual hurdles, a significant practical barrier is the skill gap, as causal machine learning draws on diverse areas of expertise. Closing the skill gap benefits from interdisciplinary collaboration and institutional support. Programs such as the Summer Institute in Computational Social Science and conferences like the International Conference on Computational Social Science have begun to foster spaces where social scientists, statisticians, and computer scientists can exchange tools, perspectives, and best practices.

Future research should further develop the epistemological foundations of causal machine learning as a deductive–inductive paradigm. We argue that causal machine learning operates through an iterative feedback loop in which theoretical assumptions guide flexible estimation, which inductively uncovers complex heterogeneities and nonlinearities, and these discoveries in turn refine or challenge the assumptions that make causal claims possible. Formalizing this cycle is an important next step: Researchers need conceptual frameworks that clarify when theory should constrain high-capacity models and when inductively uncovered patterns should prompt revisions to identification strategies. Advances in representation learning and simulation-based modeling further extend these possibilities by enabling the discovery of empirical structure that may correspond to, refine, or even generate new sociological concepts. Taken together, these developments point to the need for a richer epistemology in which deductive causal identification and inductive machine learning jointly shape the formation of theory, the design of empirical strategies, and the production of sociological knowledge.

The integration of causal machine learning into sociological research promises to deepen theoretical insight, enhance empirical flexibility, and improve methodological rigor. By uniting the deductive structure of causal identification with the inductive strength of machine learning, researchers can estimate heterogeneous effects, uncover causal mechanisms, and analyze life-course processes—all while remaining grounded in sociological theory. Emerging developments such as design-based methods for high-dimensional data and generative models for causal reasoning are opening new analytic possibilities. As these tools evolve, so too does sociology’s capacity to address complex empirical and theoretical questions in a rapidly changing methodological landscape.

7 Appendix

7.1 Appendix A: Reviewed Articles

This section includes the full list of articles reviewed for the synthesis paper.

Table A.1 Overview of sociological articles on causal inference and machine learning

No.	Authors	Title	Journal	Year	Type	Class.	Theme 1	Theme 2
1	Koch Bernard J, Sainburg Tim, Geraldo Bastias Pablo, Jiang Song, Sun Yizhou, Foster Jacob G	A primer on deep learning for causal inference	Sociological Methods & Research	2024	Methodological	Causal + ML	Others	-
2	Kaiser F, Oberwittler D, Thielmann I, Kleinke K, Greifer N	When does criminal victimization undermine generalized trust? A weighted panel analysis of the effects of crime type, frequency, and variety	Social Science Research	2024	Substantive	Causal	Time-varying	-
3	Lee B, Pescosolido BA	Misery needs company: Contextualizing the geographic and temporal link between unemployment and suicide	American Sociological Review	2024	Substantive	Causal	Quasi-experimental	-
4	Glynn AN, Rueda MR, Schuessler J	Post-instrument bias in linear models	Sociological Methods & Research	2024	Methodological	Causal	Others	-
5	Runhardt RW	Concrete counterfactual tests for process tracing: Defending an interventionist potential outcomes framework	Sociological Methods & Research	2024	Methodological	Causal	Others	-
6	Hong G, Chung HJ	Assessing the impact of the Great Recession on the transition to adulthood	Sociological Methods & Research	2024	Methodological	Causal	Others	-
7	Rubineau B, Noh S, Neblo MA, Lazer DM	Pathways of peer influence on major choice	Social Forces	2024	Substantive	Causal	Others	-
8	Mijs JJ, Usmani A	How segregation ruins inference: A sociological simulation of the inequality equilibrium	Social Forces	2024	Substantive	Causal	Others	-
9	Zhou X	Attendance, completion, and heterogeneous returns to college: A causal mediation approach	Sociological Methods & Research	2024	Methodological	Causal + ML	Mediation	-

Table A.1 (Continued)

No.	Authors	Title	Journal	Year	Type	Class.	Theme 1	Theme 2
10	Daoud A, Johansson FD	The impact of austerity on children: Uncovering effect heterogeneity by political, economic, and family factors in low- and middle-income countries	Social Science Research	2024	Substantive	Causal + ML	Heterogeneity	–
11	Lim C, Wiertz D	Civic lessons that last? Religiosity and volunteering on the way to adulthood	American Sociological Review	2024	Substantive	Causal	Time-varying	Mediation
12	Zang E, Sobel ME, Luo L	The mobility effects hypothesis: Methods and applications	Social Science Research	2023	Substantive	Causal + ML	Others	–
13	Baumgartner Michael, Ambühl Mathias	Optimizing consistency and coverage in configurational causal modeling	Sociological Methods & Research	2023	Methodological	Causal	Others	–
14	Schuessler J, Selb P	Graphical causal models for survey inference	Sociological Methods & Research	2023	Methodological	Causal	Others	–
15	Gorman E	Does schooling have lasting effects on cognitive function? Evidence from compulsory schooling laws	Demography	2023	Substantive	Causal	Quasi-experimental	–
16	Theilmann Michaela, Geldsetzer Pascal, Bärnighausen Till, Sudharsanan, Nikkil	Does early childhood BCG vaccination improve survival to midlife in a population with a low tuberculosis prevalence? Quasi-experimental evidence on nonspecific effects from 32 Swedish birth cohorts	Demography	2023	Substantive	Causal	Quasi-experimental	–
17	Potente Cecilia, Prägn Patrick, Monden Christiaan WS	Does children's education improve parental health and longevity? Causal evidence from Great Britain	Journal of Health and Social Behavior	2023	Substantive	Causal	Quasi-experimental	–
18	Tortù Costanza, Crimaldi Irene, Mealli Fabrizia, Forastiere, Laura	Estimating causal effects of multi-valued treatments accounting for network interference: Immigration policies and crime rates	Sociological Methods & Research	2023	Methodological	Causal	Others	–

Table A.1 (Continued)

No.	Authors	Title	Journal	Year	Type	Class.	Theme 1	Theme 2
19	Zhou X, Pan G	Higher Education and the Black–White earnings gap	American Sociological Review	2023	Substantive	Causal + ML	Mediation	–
20	Duxbury SW	A threatening tone: Homicide, racial threat narratives, and the historical growth of incarceration in the United States, 1926–2016	Social Forces	2023	Substantive	Causal	Mediation	–
21	O’Flaherty M, Kalucza S, Bon J	Does anyone suffer from teenage motherhood? Mental health effects of teen motherhood in Great Britain are small and homogeneous	Demography	2023	Substantive	Causal + ML	Heterogeneity	–
22	Elwert F, Pfeffer FT	The future strikes back: Using future treatments to detect and reduce hidden bias	Sociological Methods & Research	2022	Methodological	Causal	Time-varying	–
23	Lundberg I	The gap-closing estimand: A causal approach to study interventions that close disparities across social categories	Sociological Methods & Research	2022	Methodological	Causal + ML	Others	–
24	Graetz N, Boen CE, Esposito MH	Structural racism and quantitative causal inference: A life course mediation framework for decomposing racial health disparities	Journal of Health and Social Behavior	2022	Substantive	Causal	Time-varying	–
25	Wodtke GT, Ramaj S, Schachner J	Toxic neighborhoods: The effects of concentrated poverty and environmental lead contamination on early childhood development	Demography	2022	Substantive	Causal	Time-varying	–
26	Stolzenberg RM, Lindgren J	Judges as party animals: Retirement timing by federal judges and party control of judicial appointments	American Sociological Review	2022	Substantive	Causal	Quasi-experimental	–
27	Shi D, Tong X	Mitigating selection bias: A Bayesian approach to two-stage causal modeling with instrumental variables for nonnormal missing data	Sociological Methods & Research	2022	Methodological	Causal	Others	–

Table A.1 (Continued)

No.	Authors	Title	Journal	Year	Type	Class.	Theme 1	Theme 2
28	Leszczensky L, Wolbring T	How to deal with reverse causality using panel data? Recommendations for researchers based on a simulation study	Sociological Methods & Research	2022	Methodological	Causal	Quasi-experimental	–
29	Li T, Frank K	The probability of a robust inference for internal validity	Sociological Methods & Research	2022	Methodological	Causal	Others	–
30	Halpern-Manners A, Hernandez EM, Wilbur TG	Crossover effects of education on health within married couples	Journal of Health and Social Behavior	2022	Substantive	Causal	Quasi-experimental	–
31	Wei L	When the wall is broken: Rural-to-urban migration, perceived inequality, and subjective social status in China	Research in Social Stratification and Mobility	2022	Substantive	Causal	Quasi-experimental	–
32	Mark ND, Geller A, Engberg J	Adding insult to injury: Arrests reduce attendance through institutional mechanisms	Sociology of Education	2022	Substantive	Causal	Quasi-experimental	–
33	An W, Vander-Weele TJ	Opening the blackbox of treatment interference: Tracing treatment diffusion through network analysis	Sociological Methods & Research	2022	Methodological	Causal	Others	–
34	Ruppel EH, Child S, Fischer CS, Botchway M	Causal relationships between personal networks and health: A comparison of three modeling strategies	Journal of Health and Social Behavior	2022	Substantive	Causal	Others	–
35	Zhao L, Hessel P, Simon Thomas J, Beckfield J	Inequality in place: Effects of exposure to neighborhood-level economic inequality on mortality	Demography	2021	Substantive	Causal	Time-varying	–
36	Lundberg Ian, Johnson Rebecc, Stewart Brandon M	What is your estimand? Defining the target quantity connects statistical evidence to theory	American Sociological Review	2021	Substantive	Causal + ML	Others	–

Table A.1 (Continued)

No.	Authors	Title	Journal	Year	Type	Class.	Theme 1	Theme 2
37	Kim Y, Steiner PM	Gain scores revisited: A graphical models perspective	Sociological Methods & Research	2021	Methodological	Causal	Others	–
38	An W, N, Glynn A	Treatment effect deviation as an alternative to Blinder–Oaxaca decomposition for studying social inequality	Sociological Methods & Research	2021	Methodological	Causal	Others	–
39	Gonzalez-Ocantos E, LaPorte J	Process tracing and the problem of missing data	Sociological Methods & Research	2021	Methodological	Causal	Others	–
40	Bloome D, Schrage D	Covariance regression models for studying treatment effect heterogeneity across one or more outcomes: Understanding how treatments shape inequality	Sociological Methods & Research	2021	Methodological	Causal + ML	Heterogeneity	–
41	Brand JE, Xu J, Koch B, Geraldo P	Uncovering sociological effect heterogeneity using tree-based machine learning	Sociological Methodology	2021	Methodological	Causal + ML	Heterogeneity	–
42	Quintana R	The structure of academic achievement: searching for proximal mechanisms using causal discovery algorithms	Sociological Methods & Research	2020	Methodological	Causal + ML	Others	–
43	Wodtke GT	Regression-based adjustment for time-varying confounders	Sociological Methods & research	2020	Methodological	Causal	Time-varying	–
44	Barban N, de Luna X, Lundholm E, Svensson I, Billari FC	Causal effects of the timing of life-course events: Age at retirement and subsequent health	Sociological Methods & Research	2020	Methodological	Causal	Quasi-experimental	–
45	Rangel MA, Nobles J, Hamoudi A	Brazil's missing infants: Zika risk changes reproductive behavior	Demography	2020	Substantive	Causal	Quasi-experimental	–

Table A.1 (Continued)

No.	Authors	Title	Journal	Year	Type	Class.	Theme 1	Theme 2
46	Torrats-Espinoso G	Crime and inequality in academic achievement across school districts in the United States	Demography	2020	Substantive	Causal	Quasi-experimental	–
47	Wenz SE, Hoeng K	Ethnic and social class discrimination in education: Experimental evidence from Germany	Research in Social Stratification and Mobility	2020	Substantive	Causal	Quasi-experimental	–
48	Perkins KL	Changes in household composition and children's educational attainment	Demography	2019	Substantive	Causal	Time-varying	–
49	Fosse E, Winship C	Bounding analyses of age-period-cohort effects	Demography	2019	Substantive	Causal	Others	–
50	Sohn H, Lee SW	Causal impact of having a college degree on women's fertility: Evidence from regression kink designs	Demography	2019	Substantive	Causal	Others	–
51	Zhou X	Equalization or selection? Reassessing the meritocratic power of a college degree in intergenerational income mobility	American Sociological Review	2019	Substantive	Causal	Mediation	–
52	Becker Rolf, Möser Sara, Glauser, David	Cash vs. vouchers vs. gifts in web surveys of a mature panel study—Main effects in a long-term incentives experiment across three panel waves	Social Science Research	2019	Substantive	Causal	Others	–
53	Dafoe A	Nonparametric identification of causal effects under temporal dependence	Sociological methods & research	2018	Methodological	Causal	Time-varying	–
54	Hicks AL, Handcock MS, Sastry N, Pebley AR	Sequential neighborhood effects: The effect of long-term exposure to concentrated disadvantage on children's reading and math test scores	Demography	2018	Substantive	Causal	Time-varying	–

Table A.1 (Continued)

No.	Authors	Title	Journal	Year	Type	Class.	Theme 1	Theme 2
55	Aksoy O, Billari FC	Political Islam, marriage, and fertility: Evidence from a natural experiment	American Journal of sociology	2018	Substantive	Causal	Quasi-experimental	–
56	Torche F	Prenatal exposure to an acute stressor and children's cognitive outcomes	Demography	2018	Substantive	Causal	Quasi-experimental	–
57	Daniel A, Watermann R	The role of perceived benefits, costs, and probability of success in students' plans for higher education. A quasi-experimental test of rational choice theory	European Sociological Review	2018	Substantive	Causal	Quasi-experimental	–
58	Basse GW, Airolidi EM	Limitations of design-based causal inference and A/B testing under arbitrary and network interference	Sociological methodology	2018	Methodological	Causal	Others	–
59	An W	Causal inference with networked treatment diffusion	Sociological Methodology	2018	Methodological	Causal	Others	–
60	Wodtke GT, Almirall D	Estimating moderated causal effects with time-varying treatments and time-varying moderators: Structural nested mean models and regression with residuals	Sociological Methodology	2017	Methodological	Causal	Time-varying	–
61	Steiner PM, Kim Y, Hall CE, Su D	Graphical models for quasi-experimental designs	Sociological Methods & Research	2017	Methodological	Causal	Others	–
62	An W, Winship C	Causal inference in panel data with application to estimating race-of-interviewer effects in the General Social Survey	Sociological Methods & Research	2017	Methodological	Causal	Others	–
63	Bijlsma MJ, Daniel RM, Janssen F, De Stavola BL	An assessment and extension of the mechanism-based approach to the identification of age-period-cohort models	Demography	2017	Substantive	Causal	Others	–

Table A.1 (Continued)

No.	Authors	Title	Journal	Year	Type	Class.	Theme 1	Theme 2
64	Zheng H	Why does college education matter? Unveiling the contributions of selection factors	Social Science Research	2017	Substantive	Causal	Quasi-experimental	—
65	Lawrence M, Breen R	<i>And Their Children after Them? The Effect of College on Educational Reproduction</i>	American journal of sociology	2016	Substantive	Causal	Others	—
66	Alvarado SE	Neighborhood disadvantage and obesity across childhood and adolescence: Evidence from the NLSY children and young adults cohort (1986–2010)	Social Science Research	2016	Substantive	Causal	Quasi-experimental	—
67	Minkoff D	The payoffs of organizational membership for political activism in established democracies	American Journal of Sociology	2016	Substantive	Causal	Quasi-experimental	—
68	Gamoran A, Barfels S, Collares AC	Does racial isolation in school lead to long-term disadvantages? Labor market consequences of high school racial composition	American Journal of Sociology	2016	Substantive	Causal	Quasi-experimental	—
69	Legewie J	Racial Profiling and Use of Force in Police Stops: How Local Events Trigger Periods of Increased Discrimination	American Journal of Sociology	2016	Substantive	Causal	Quasi-experimental	—
70	Roth T, Siegert M	Does the selectivity of an educational system affect social inequality in educational attainment? Empirical findings for the transition from primary to secondary level in Germany	European Sociological Review	2016	Substantive	Causal	Quasi-experimental	—
71	Elías Alvarado	Delayed disadvantage: Neighborhood context and child development	Social Forces	2016	Substantive	Causal	Quasi-experimental	—
72	Pedulla, David S	Penalized or protected? Gender and the consequences of non-standard and mismatched employment histories	American Sociological Review	2016	Substantive	Causal	Mediation	—

Table A.1 (Continued)

No.	Authors	Title	Journal	Year	Type	Class.	Theme 1	Theme 2
73	Wodtke Geoffrey T, Harding David J, Elwert Felix	Neighborhood effect heterogeneity by family income and developmental period	American Journal of Sociology	2016	Substantive	Causal	Time- varying	Hetero- geneity
74	Ferguson JP	The control of managerial discretion: Evi- dence from unionization's impact on employ- ment segregation	American Journal of Sociology	2015	Substantive	Causal	Quasi- experi- mental	–
75	An W	Instrumental variables estimates of peer effects in social networks	Social Science Research	2015	Substantive	Causal	Others	Quasi- experi- mental
76	Tinkler JE, Bunker Whittington K, Ku MC, Davies AR	Gender and venture capital decision-making: The effects of technical background and social capital on entrepreneurial evaluations	Social Science Research	2015	Substantive	Causal	Quasi- experi- mental	–
77	Yamaguchi K	Decomposition of gender or racial inequality with endogenous intervening covariates: An extension of the DiNardo-Fortin-Lemieux method	Sociological Methodology	2015	Methodolo- gical	Causal	Others	–
78	Lee D, McLanahan S	Family structure transitions and child devel- opment: Instability, selection, and population heterogeneity: Instability, selection, and pop- ulation heterogeneity	American Sociological Review	2015	Substantive	Causal	Time- varying	Hetero- geneity
79	Caudillo ML, Torche F	Exposure to local homicides and early educa- tional achievement in Mexico	Sociology of Education	2014	Substantive	Causal	Quasi- experi- mental	–

7.2 Appendix B: From Estimands to Identification—A Graphical Approach

This appendix summarizes key empirical estimands used across three major domains of causal inference—treatment effect heterogeneity, causal mechanism analysis, and time-varying processes—as discussed in the main text. These estimands provide the quantitative targets for identifying and estimating causal effects. Within mechanism analysis, we focus on mediation estimands (e.g., natural direct and indirect effects), which offer a formal and widely adopted foundation for decomposing total effects into direct and indirect effects. While other estimands—such as causal decomposition estimands for analyzing group disparities—have recently emerged, we do not include them here, as they are more context specific and less standardized across empirical applications.

7.2.1 Appendix B.1. Causal Estimands by Each Theme

Table B.1 Causal effect heterogeneity estimands

Estimand	Description
$E[Y(1) - Y(0)]$	<i>Average treatment effect (ATE)</i> This estimand represents the expected difference in potential outcomes across the entire population
$E[Y(1) - Y(0) \mid A = 1]$	<i>Average treatment effect among treated (ATT)</i> This estimand represents the expected treatment effect specifically for those who received the treatment
$E[Y(1) - Y(0) \mid X = x]$	<i>Conditional average treatment effect (CATE)</i> This estimand represents the expected treatment effect for units with specific covariate values $X = x$

Table B.2 Causal mediation estimands

Estimand	Description
<i>Total effect</i>	
$E[Y(1) - Y(0)] = E[Y(1, M(1))] - E[Y(0, M(0))]$	Average treatment effect
<i>Indirect effect</i>	
$E[Y(t, M(1))] - E[Y(t, M(0))]$	Average causal mediation effect
$[Y(t, M(1))] - [Y(t, M(0))]$	Natural indirect effect
$[Y(0, M(1))] - [Y(0, M(0))]$	Pure indirect effect
$[Y(1, M(1))] - [Y(1, M(0))]$	Total indirect effect
<i>Direct effect</i>	
$E[Y(1, M(t))] - E[Y(0, M(t))]$	Average direct effect
$[Y(1, M(t))] - [Y(0, M(t))]$	Natural direct effect
$[Y(1, M(0))] - [Y(0, M(0))]$	Pure direct effect
$[Y(1, M(1))] - [Y(0, M(1))]$	Total direct effect

Terminology follows standard usage in the mediation literature: natural direct and indirect effects (Pearl 2001), pure and total effects (VanderWeele 2014), and average causal mediation/direct effects such as average causal mediation effect and average direct effect (Imai et al. 2010)

Table B.3 Key estimand for time-varying causal inference

Estimand	Description
$E[Y(\bar{a}^{(1)})] - E[Y(\bar{a}^{(0)})]$	Average treatment effect of treatment trajectories. This estimand represents the total effect of a treatment sequence. Here, $\bar{a}^{(1)}$ represents a sequence of treatment interventions, while $\bar{a}^{(0)}$ represents the control scenario, often reflecting the natural course where no intervention is applied The difference captures the expected change in potential outcome when following the treatment sequence versus the control sequence. This estimand is particularly useful in longitudinal settings in which treatments are administered over time, allowing researchers to assess the cumulative impact of interventions

7.2.2 Appendix B.2. Causal Identification with Directed Acyclic Graphs

Directed acyclic graphs (DAGs) offer a graphical language for encoding theoretical assumptions about causal relationships; DAGs help researchers visualize and evaluate causal pathways, clarify assumptions, and thereby formalize theoretical mechanisms in social science (Elwert 2013; VanderWeele 2015). Pearl (2000) formalized these ideas, laying the foundation for structural causal modeling. In a DAG, nodes represent variables and are connected by directed arrows ($\rightarrow$) representing direct causal relationships. These arrows are “directed” (i.e., they only go one way), and the graph is “acyclic” (i.e., no feedback loops).

A key advantage of DAGs is their ability to diagnose confounding. Estimating causal effects relies on identification assumptions: conditional exchangeability, positivity, consistency, and the stable unit treatment value assumption (see Hernán and Robins [2020] for a review). Among these, conditional exchangeability is especially relevant in observational studies, where treatment is not randomly assigned. It implies that, after adjusting for a set of covariates, treatment assignment is independent of potential outcomes. Directed acyclic graphs help assess whether this assumption is plausible by making backdoor paths—noncausal associations due to shared causes—visually explicit. This guides researchers in selecting covariates for adjustment (Cinelli et al. 2024; Elwert 2013; Elwert and Winship 2014).

To improve the plausibility of exchangeability, researchers typically adjust for common causes of both treatment and outcome. This idea is formalized in the backdoor criterion, which states that a set of variables satisfies the criterion if (1) it blocks all backdoor paths between the treatment and the outcome, and (2) it does not include any descendant of the treatment. When these conditions are met, adjusting for the set allows for an unbiased estimate of the causal effect (Pearl et al. 2016). For examples in sociology, see Sharkey and Elwert (2011).

Figure B.1 illustrates a DAG based on Blau and Duncan’s status attainment model (Blau and Duncan 1967, p. 170) showing how a father’s occupation influences a son’s occupation. The original path diagrams included cyclical arrows indicating correlations among variables. In contrast, Fig. B.1 modifies the original by using acyclic, one-directional arrows to represent causal paths.

Figure B.2 shows the DAG after applying the backdoor criterion to this example to estimate the total effects of the father’s occupation on the son’s occupation. A key backdoor path runs through father’s education, which affects both father’s

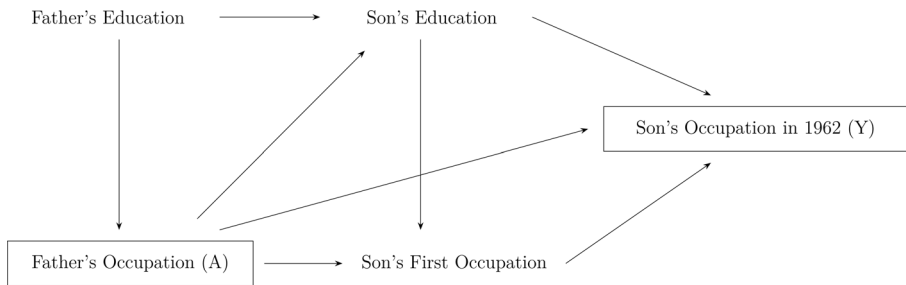


Fig. B.1 Causal Pathways Between Father's Occupation and Son's Occupation in 1962 Based on the Blau and Duncan Status Attainment Model

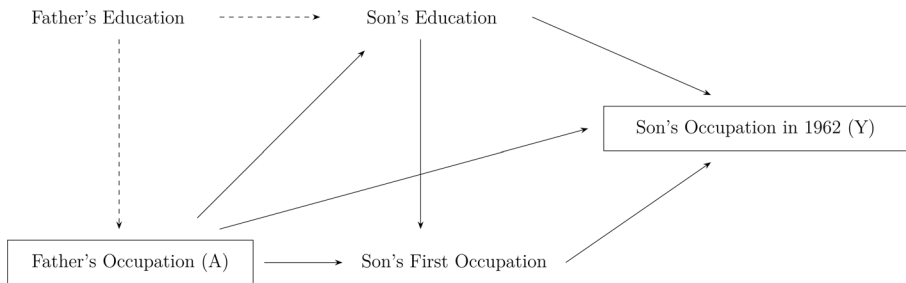


Fig. B.2 Applying the Backdoor Criterion to Identify the Total Effect of Father's Occupation on Son's Occupation in 1962

occupation and son's education—and in turn, son's occupation. To block this path, we adjust for the father's education, as shown by the dotted line. In contrast, son's education and son's first occupation are mediators, not confounders—adjusting for them would introduce post-treatment bias. Therefore, the sufficient adjustment set includes only the father's education, allowing for an unbiased estimate of the total causal effect.

Acknowledgements The authors benefited from facilities and resources provided by the California Center for Population Research at UCLA, which receives core support (P2C-HD041022) from the Eunice Kennedy Shriver National Institute of Child Health and Human Development. The authors also received helpful feedback from Heinz Leitgöb, Nico Sonntag, and Ramona Roller, as well as from the members of the Inequality Data Science Lab and participants of the authors' workshop for the special issue of *Kölner Zeitschrift für Soziologie & Sozialpsychologie (KZfSS)* on explanation and causality in the social sciences. The ideas expressed herein are those of the authors.

Conflict of interest N. Jeon and J.E. Brand declare that they have no competing interests.

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